Mizer model

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Loading library

Load the packages required to design mizer model and produce figures.

```
library(mizer)
library(mizerExperimental)
library(tidyverse)
```

Setting initial parameters

```
params_single <- newSingleSpeciesParams()
params_initial <- setResource(params_single,
                      resource_dynamics = "resource_semichemostat")
params_double <- params_initial
params_double <- setResource(params_double,
                             resource_level = 0.1)

initialN(params_double) <- 2*initialN(params_double)
initialNResource(params_double) <- initialNResource(params_double)/2

repro_prop(params_double) <- repro_prop(params_double)/3
```

Implementing fishing gear

```
gear_params(params_double) <- data.frame(
  gear = "gear",
  species = "Target species",
  catchability = 0.3,
```

```r
  sel_func = "sigmoid_weight",
  sigmoidal_weight = 15,
  sigmoidal_sigma = 5)

params_double <- setFishing(params_double, gear_params = gear_params)

gear_params(params_double)
```

```
                     gear        species catchability       sel_func
Target species, gear gear Target species          0.3 sigmoid_weight
                     sigmoidal_weight sigmoidal_sigma
Target species, gear               15               5
```

Simulating ecosystem behaviour

```r
sim_double <- project(params_double, t_max = 15, effort = 1)
```

Extract and plot yield

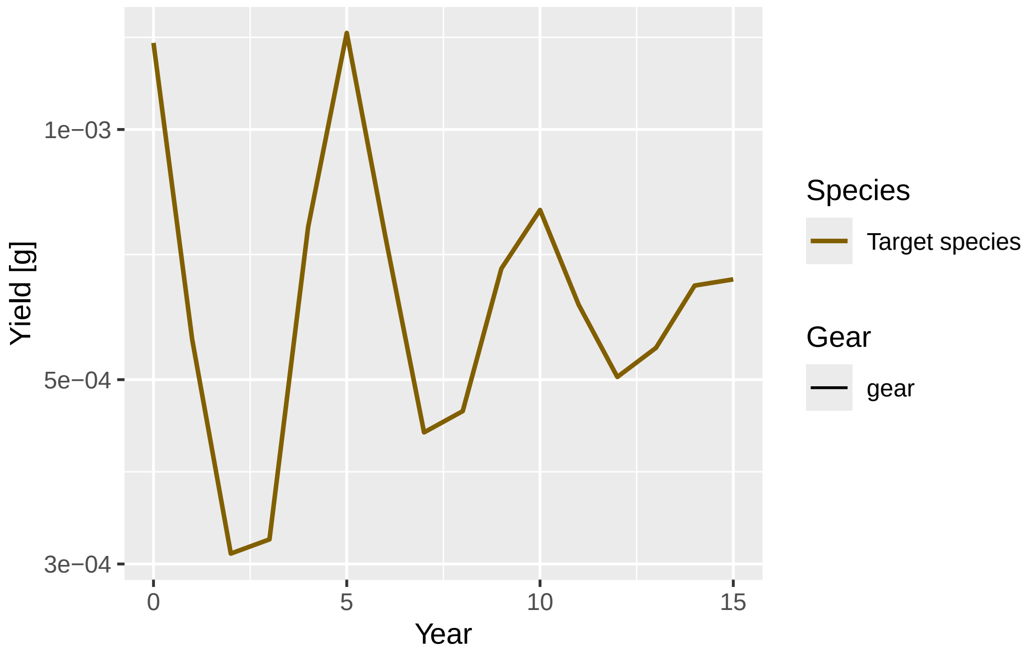
```r
getYieldGear(sim_double)
```

```
, , sp = Target species

    gear
time         gear
  0  0.0012712274
  1  0.0005590655
  2  0.0003087767
  3  0.0003211698
  4  0.0007631445
  5  0.0013068563
  6  0.0007427161
  7  0.0004320280
  8  0.0004582261
  9  0.0006801194
  10 0.0008002619
  11 0.0006152881
  12 0.0005037359
```

```
## 13 0.0005460477
## 14 0.0006489137
## 15 0.0006601905
```

```
plotYieldGear(sim_double)
```



Locally reduce resource parameters

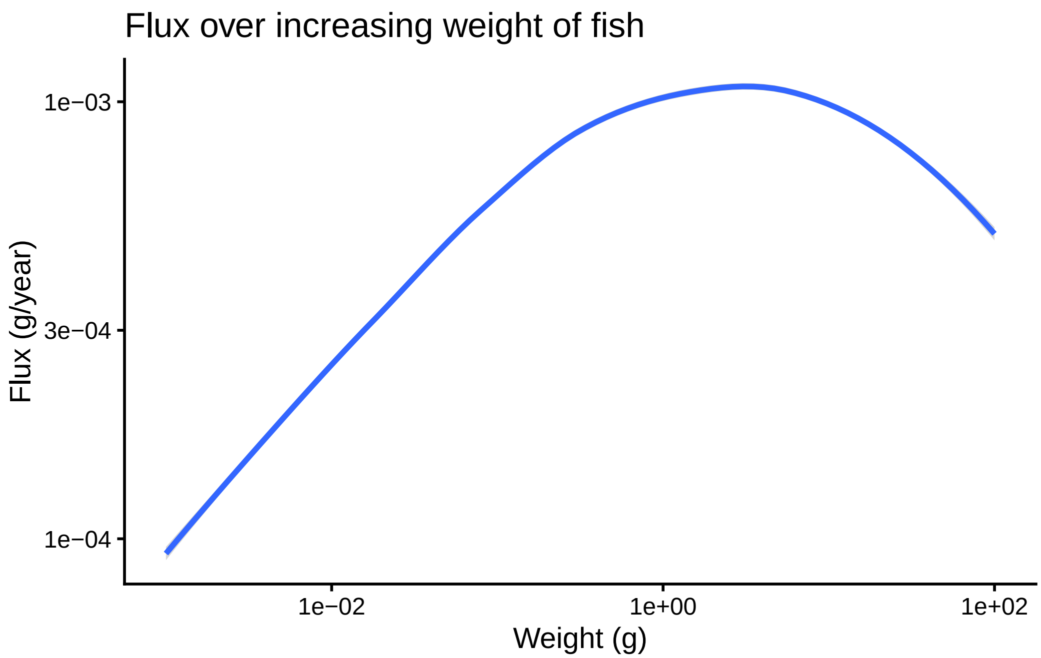
Calculate current flux

```
N <- finalN(sim_double)["Target species", , drop = TRUE]
w <- w(params_double)

E_growth <- getEGrowth(params_double)["Target species", , drop = TRUE]
gr <- w * E_growth
flux <- gr * N
```

Plot initial flux without local reductions

```
initial_flux_data <- data.frame(Weight = w,
                                Flux = flux)
initial_flux_plot <- ggplot(initial_flux_data,
                            aes(x = Weight, y = Flux)) +
  geom_smooth() +
  scale_x_log10() +
  scale_y_log10() +
  labs(
    x = paste0("Weight (g)"),
    y = paste0("Flux (g/year)"),
    title = "Flux over increasing weight of fish") +
  theme_classic()
initial_flux_plot
```



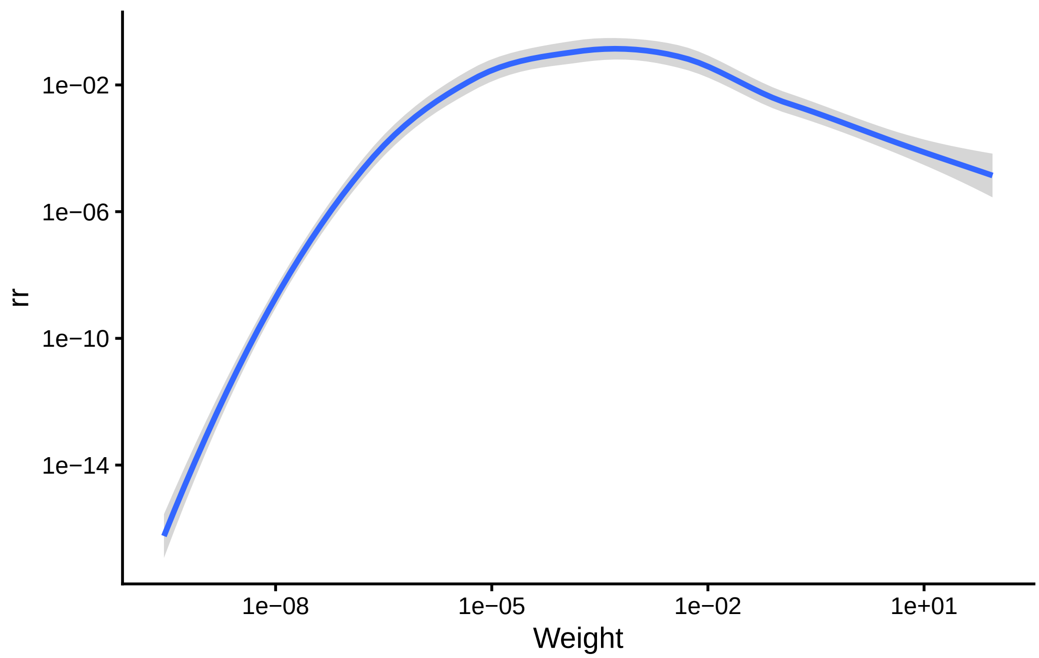
Locally reduce resource rate

```
rr <- resource_rate(params_double)
w_full <- w_full(params_double)
rr[230:250] <- rr[230:250] / 1000000

rr_data <- data.frame(
```

```
  Weight = params_double@w_full,
  rr = rr)

rr_plot <- ggplot(rr_data,
                  aes(x = Weight, y = rr)) +
  geom_smooth() +
  scale_x_log10() +
  scale_y_log10() +
  theme_classic()
rr_plot
```

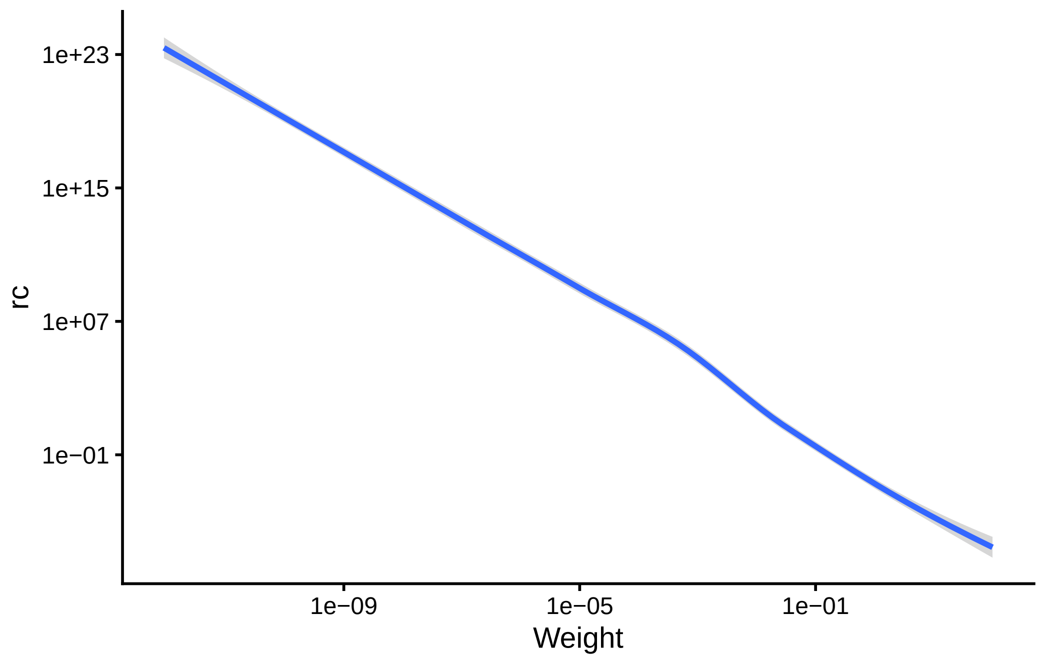


Locally reduce resource capacity

```
rc <- resource_capacity(params_double)
rc[230:250] <- rc[230:250] / 1000000

rc_data <- data.frame(
  Weight = params_double@w_full,
  rc = rc
)
```

```
rc_plot <- ggplot(rc_data,
                  aes(x = Weight, y = rc)) +
  geom_smooth() +
  scale_x_log10() +
  scale_y_log10() +
  theme_classic()
rc_plot
```



Set new parameters with local resource parameter reductions

```
params_reduced <- setResource(params_double,
                              resource_capacity = rc,
                              resource_rate = rr,
                              balance = FALSE)
```

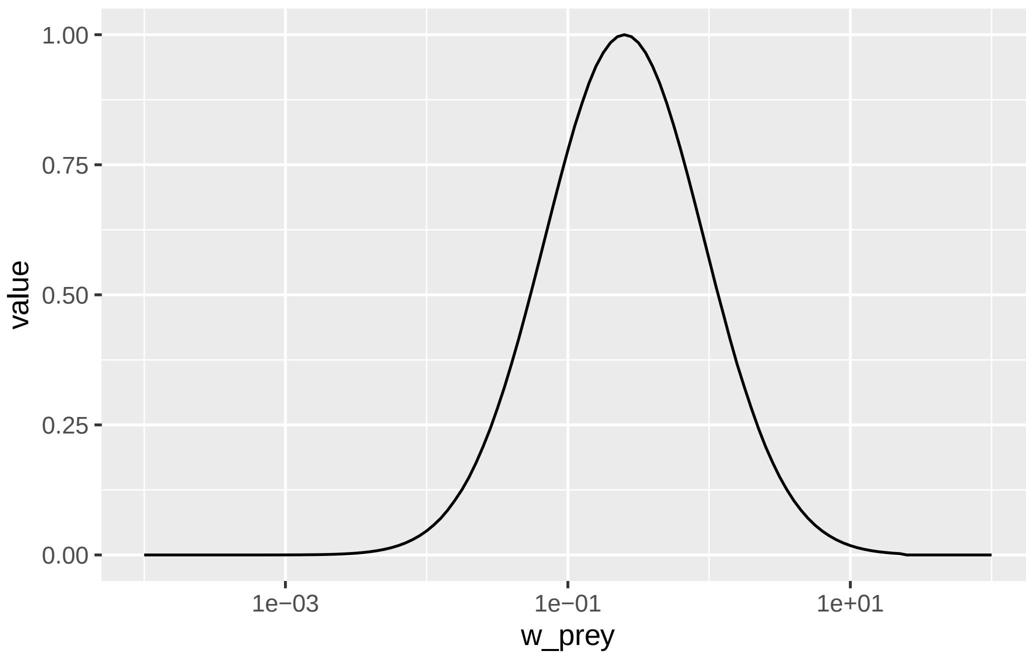
Narrow predation kernel

```
pred_kernel <- getPredKernel(params_reduced)
```

```r
pred_kernel_reduced <- pred_kernel[, 89, , drop = FALSE]

ggplot(melt(pred_kernel_reduced)) +
  geom_line(aes(x = w_prey, y = value)) +
  scale_x_log10(limits = c(1e-4, 100))
```

```
Warning: Removed 161 rows containing missing values or values outside the scale range
(`geom_line()`).
```




```r
select(species_params(params_reduced), beta, sigma)
```

```
               beta sigma
Target species  100   1.3
```

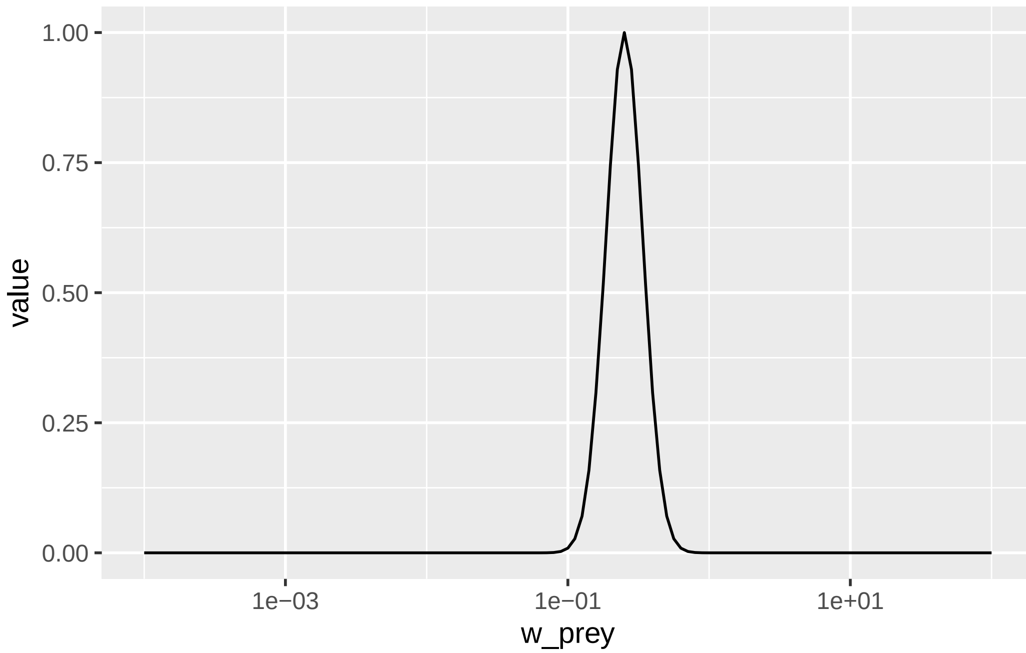
```r
params <- params_reduced

given_species_params(params)$sigma <- 0.3
given_species_params(params)$beta <- 100

getPredKernel(params)[, 89, , drop = FALSE] %>%
```

```
  melt() %>%
  ggplot() +
  geom_line(aes(x = w_prey, y = value)) +
  scale_x_log10(limits = c(1e-4, 100))
```

```
Warning: Removed 161 rows containing missing values or values outside the scale range
(`geom_line()`).
```



Calculate flux after local reductions in resource parameters

```
sim_reduced <- project(params, t_max = 15, effort = 1)

N_reduced <- finalN(sim_reduced)["Target species", , drop = TRUE]
w <- w(params_reduced)

E_growth_reduced <- getEGrowth(params)["Target species", , drop = TRUE]
grr <- w * E_growth_reduced
flux_reduced <- grr * N_reduced
```

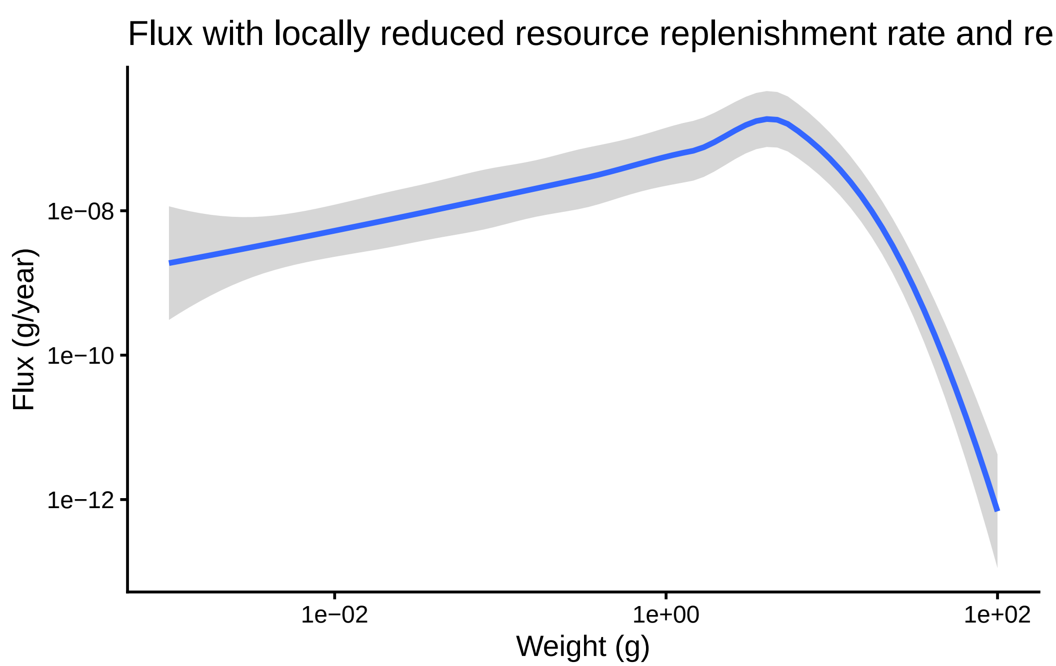
```
reduced_flux_data <- data.frame(Weight = w,
                                Flux = flux_reduced)
```

Plot on a log-log axis

```
reduced_flux_log_plot <- ggplot(reduced_flux_data,
                                aes(x = Weight, y = Flux)) +
  geom_smooth() +
  scale_y_log10() +
  scale_x_log10() +
  labs(
    x = paste0("Weight (g)"),
    y = paste0("Flux (g/year)"),
    title = "Flux with locally reduced resource replenishment rate and resource capacity") +
  theme_classic()

reduced_flux_log_plot
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

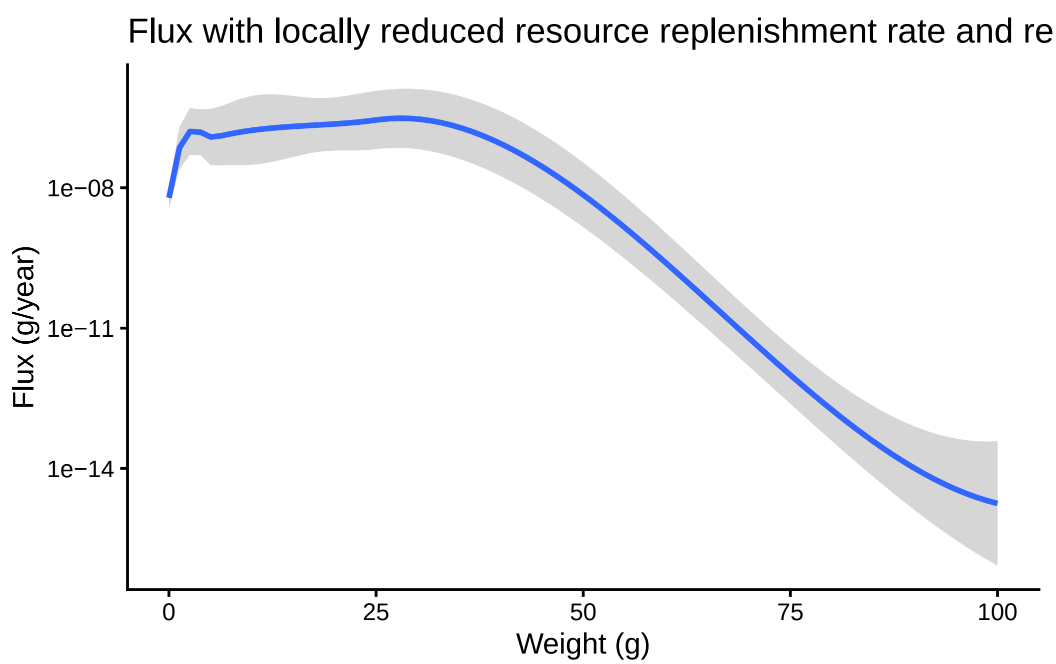


Plot on log y-axis

```
reduced_flux_plot <- ggplot(reduced_flux_data,
                            aes(x = Weight, y = Flux)) +
  geom_smooth() +
  scale_y_log10() +
  labs(
    x = paste0("Weight (g)"),
    y = paste0("Flux (g/year)"),
    title = "Flux with locally reduced resource replenishment rate and resource capacity") +
  theme_classic()

reduced_flux_plot
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



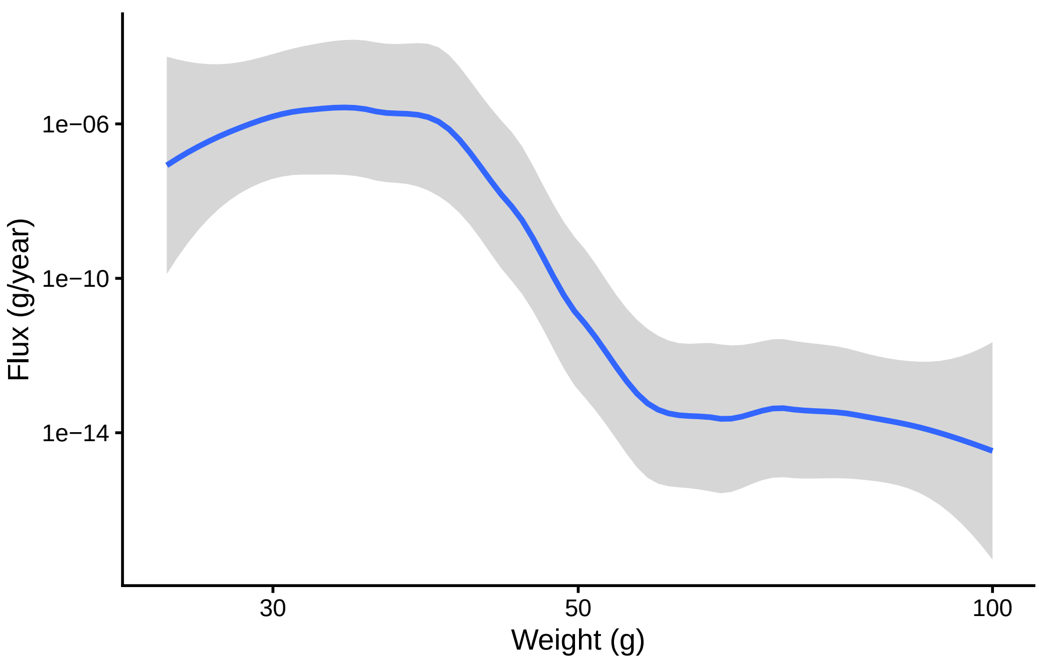
Flux over a limited weight range

```
limited_flux_data <- data.frame(Weight = w[89:101],
                                Flux = flux_reduced[89:101])
```

```
limited_flux_log_plot <- ggplot(limited_flux_data,
                                aes(x = Weight, y = Flux)) +
  geom_smooth() +
  scale_y_log10() +
  scale_x_log10(limits = c(25, 100)) +
  labs(
    x = paste0("Weight (g)"),
    y = paste0("Flux (g/year)")) +
  theme_classic()

limited_flux_log_plot
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

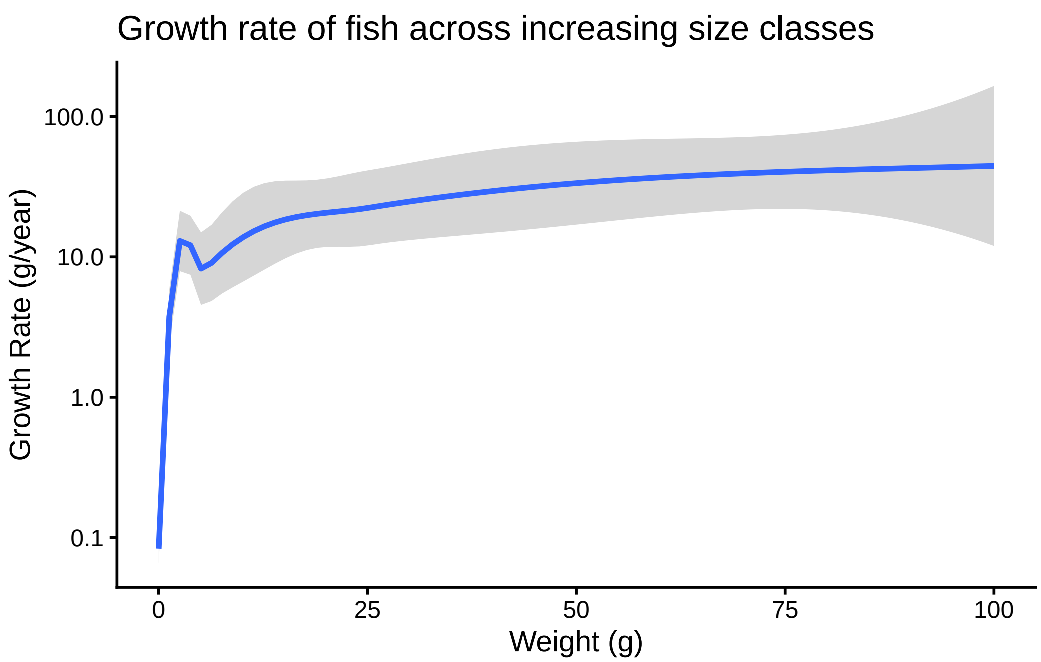


Plot growth rate against weight

```
growth_data <- data.frame(growth = E_growth_reduced,
                          weight = w)
growth_plot <- ggplot(growth_data,
```

```r
      aes(x = weight,
          y = growth)) +
  geom_smooth() +
  scale_y_log10() +
  labs(x = paste0("Weight (g)"),
       y = paste0("Growth Rate (g/year)"),
       title = "Growth rate of fish across increasing size classes") +
  theme_classic()
growth_plot
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Plot yield

```r
yield <- getYield(sim_reduced)
time <- getTimes(sim_reduced)

yield_data <- data.frame(Time = time,
                         Yield = yield)
```

```
yield_graph <- ggplot(yield_data,
                      aes(x = Time, y = Target.species)) +
  geom_line() +
  scale_y_log10() +
  labs(x = paste0("Weight (year)"),
       y = paste0("Yield (g/year)") ) +
  theme_classic()
  yield_graph
```